

1. Ray Optics for Imaging

1.1 Light as Straight Ray

Optics explores the characteristics of light and its interactions with matter. It includes several useful descriptions of light, including ray optics, wave optics, and quantum optics. Ray optics is the simplest of these models: it represents light as straight rays that show the direction of propagation. Although this representation is approximate, especially compared with wave and quantum optics, it remains a foundational tool for optical engineering and imaging-system design.

The straight-ray approximation is especially useful for understanding how light is focused by a lens. The main properties of an ideal thin lens are summarized in Figure 1.1:

1. A light ray striking the lens perpendicularly is deflected so that it passes through the lens focal point (orange rays in Figure 1.1).
2. A light ray passing through the center of the lens is not deflected (green rays in Figure 1.1).
3. Light propagation is reciprocal: if rays emitted from a source on one side of the lens converge to a point on the other side, then rays emitted from that convergence point and directed back toward the lens retrace the original paths.

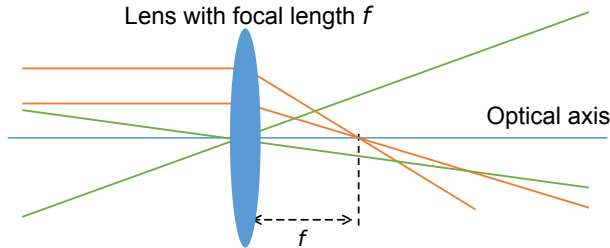


Figure 1.1. Major properties of an ideal lens.

Consider a rod whose two ends emit diverging light, positioned at a distance z_{samp} from an ideal lens. This distance is called the sample distance (Figure 1.2). Among the many rays emitted from the upper end of the rod, one ray that is parallel to the optical axis is deflected through the focal point, located

one focal length f from the lens. Another ray passes through the center of the lens and is not deflected. These two characteristic rays converge at a distance z_{img} , called the image distance. Other rays emitted from the same point on the rod also converge at this location. Similarly, rays emitted from the lower end of the rod converge at the same image distance but at a different transverse position.

If a sheet of paper is placed parallel to the lens at z_{img} , two focused spots appear at the convergence points. If the rod emits light continuously, like a glow stick, the focused spots form a line image on the paper. Ideally, all rays emitted from any point in the sample plane, located z_{samp} from the lens, converge to a corresponding point in the image plane, located z_{img} from the lens. These two planes are therefore called **conjugate planes**.

The image distance is governed by the thin-lens equation:

$$\frac{1}{z_{\text{samp}}} + \frac{1}{z_{\text{img}}} = \frac{1}{f} \quad (1.1)$$

The magnification is the ratio of the length of the projected image to the length of the original object:

$$m = \frac{z_{\text{img}}}{z_{\text{samp}}} \quad (1.2)$$

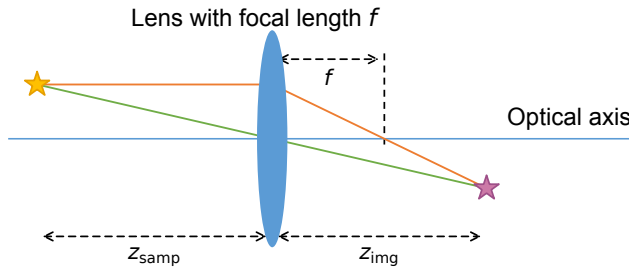


Figure 1.2. Image generated by an ideal thin lens.

Example 1.1: Simple single-lens camera

Consider a camera with a single thin lens with focal length $f = 100$ mm.

- (a) To capture a focused image of a subject 150 mm from the lens, determine the required distance between the lens and the camera sensor. Calculate the magnification.
- (b) To photograph a distant landscape, where the sample distance can be approximated as infinity, determine the required distance between the lens and the camera sensor. Calculate the magnification.